

Ismail Ahmed

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GitHub

LinkedIn

Portfolio

EDUCATION

University of Toronto

Sep. 2020 – Dec. 2025

Honours Bachelor of Science in **Computer Science**

Toronto, ON

- **Leadership:** Director at **UofT AI**; chaired technical panels with industry experts and managed a community of 200+ student developers to bridge the gap between academia and industry AI trends.
- **Relevant Coursework:** Machine Learning, Deep Learning, Operating Systems, Database Systems, Web Security.

EXPERIENCE

Lifomation

Jan. 2024 – Dec. 2024

Software Developer

Toronto, ON

- Architected an enterprise **Retrieval-Augmented Generation (RAG)** system using **FAISS** and **OpenAI Embeddings**, enabling semantic search across 10,000+ docs with 90%+ retrieval precision.
- Engineered high-throughput ingestion pipelines in **Python**, optimizing vector database indexing which reduced query latency by **45%** and improved system response times for 500+ daily active users.
- Designed **MLOps monitoring** dashboards to track model drift and retrieval accuracy in real-time.

LangChain-AI (Open Source Contributor)

Sep. 2024 – Dec. 2024

Software Developer

Remote

- Authored and upstreamed a **Dropbox document loader** module for the **LangChain.js** ecosystem, providing seamless cloud storage integration for the global developer community (30k+ stars on GitHub).
- Implemented robust error-handling and recursive file-parsing logic in **TypeScript**, ensuring stable data streaming for Large Language Model (LLM) ingestion pipelines.

University of Toronto

Jan. 2025 – May. 2025

Machine Learning Researcher

Toronto, ON

- Spearheaded research into **Generative Modeling**; trained **WGANs** and **DDM-CGANs** to synthesize financial time-series data, achieving a **98% distribution match** against historical tail-risk data.
- Quantified model reliability using **Monte Carlo simulations** and **Frobenius norms** for stress-testing.

Maple Bit Services

May. 2022 – Aug. 2022

Software Engineer Intern

Toronto, ON

- Refactored legacy backend modules into **FastAPI microservices**, utilizing **Asynchronous Programming** to boost concurrent transaction processing capacity by **35%**.
- Optimized **PostgreSQL** queries and schema designs, reducing data retrieval times for analytics by 20%.

SELECTED PROJECTS

DevSync Real-time Collaborative Workspace | React, Node.js, Socket.io, Redis

- Developed a full-stack collaborative platform with **real-time synchronization** using **WebSockets**, allowing multiple users to edit code and documents simultaneously with **< 50ms** latency.
- Implemented **Redis-based caching** for session management and **JWT authentication** for secure access.
- Built a responsive UI with **Tailwind CSS** and integrated a custom Markdown compiler for live previews.

VisionEdge | Python, TensorFlow, OpenCV, Flask, Docker

- Developed a **lightweight CNN** optimized for edge devices, achieving real-time inference (30+ FPS) and a **92% mAP**. Containerized with **Docker** for seamless cross-platform deployment.

TECHNICAL SKILLS

Languages: Python (Expert), C++, Java, TypeScript, SQL, Bash, R, HTML/CSS

ML/AI: PyTorch, TensorFlow, Scikit-learn, FAISS, Transformers, RAG, GANs, Computer Vision, NLP

Frameworks/Web: React, Node.js, FastAPI, Flask, LangChain, Socket.io, Tailwind CSS, PostgreSQL

Tools: Docker, Kubernetes, AWS Sagemaker, Git, CI/CD (GitHub Actions), Redis, Azure ML